CS 310 : Automata Theory

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- $\sim_L \subseteq \Sigma^* \times \Sigma^*$ is an equivalence relation on words
- Consider the equivalence classes of $\sim_L$
- When can $\sim_L$ have only finitely many equivalence classes?

Recall : Properties of $\sim_L$

- $\sim_L$ is a right congruence : that is, for any $a \in \Sigma$, $x \sim_L y \Rightarrow xa \sim_L ya$
- $\sim_L$ refines L : that is, $x \sim_L y \Rightarrow x \in L \Leftrightarrow y \in L$

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Recall : Myhill-Nerode relations

An equivalence relation on Σ^* is called a *Myhill-Nerode* relation for $L \subseteq \Sigma^*$ if it is a right congruence refining L, and has finitely many equivalence classes. The relation was proposed by John Myhill and Anil Nerode.

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If L is regular, then there exists a Myhill-Nerode relation $\sim_A$ for L defined from the DFA A for L.

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- Let $A = (Q, \Sigma, \delta, q_0, F)$ be the DFA with no inaccessible states such that $L = L(A)$.
- Define $x \sim_A y$ iff $\hat{\delta}(q_0, x) = \hat{\delta}(q_0, y)$.
- It can be seen that $\sim_A$ is a right congruence refining L.
- $[x] = \{y \mid \hat{\delta}(q_0, x) = \hat{\delta}(q_0, y)\}$. There is one equivalence class corresponding to each state in Q. Hence, finitely many classes.
- Hence, $\sim_A$ is Myhill-Nerode.

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- We know $\sim$ is a right congruence refining L, with finitely many equivalence classes $[x]$.
- From $\sim$, define a DFA $A_\sim = (Q, \Sigma, \delta, s, F)$ where
  $Q = \{[x] \mid x \in \Sigma^*\}$
  $s = [\epsilon]$
  $F = \{[x] \mid x \in L\}$
  $\delta([x], a) = [xa]$.
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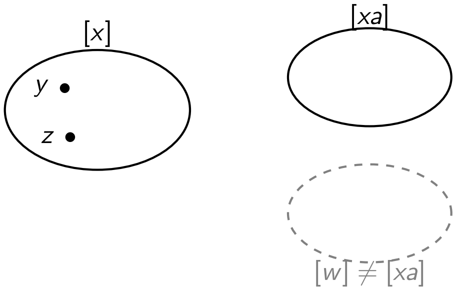

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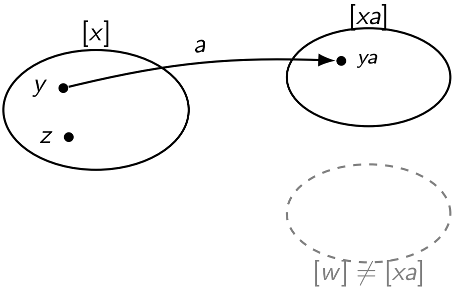

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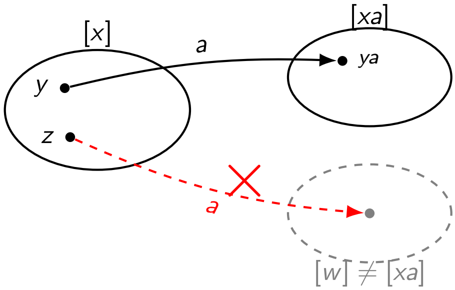

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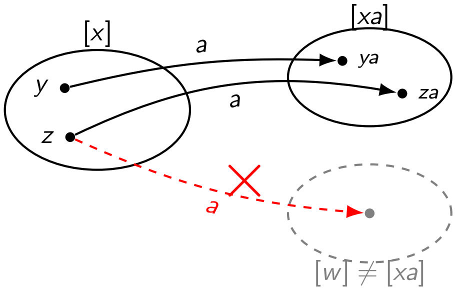

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Connecting Myhill-Nerode Automata and Minimal Automata

Preliminaries

- A relation $\sim_1$ is said to refine another relation $\sim_2$ if $\sim_1 \subseteq \sim_2$ when considered as sets of ordered pairs.
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- Any set U has a finest and coarsest equivalence relation :finest is the identity and coarsest is the universal relation $\{(x, y) \mid x, y \in U\}$.

Preliminaries

Let L be some language.

Recall : for $x, y \in \Sigma^*$, $x \sim_L y$ iff $\forall z \in \Sigma^* \, (xz \in L \Leftrightarrow yz \in L)$.

$\sim_L$ is a right congruence refining L and is the coarsest such relation.

- We already know $\sim_L$ is a right congruence refining L.
- Show that $\sim_L$ is coarsest.

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- Let $\equiv$ be any right congruence on Σ^* refining L : we show $\equiv$ refines $\sim_L$
  - $x \equiv y \Rightarrow \forall z(xz \equiv yz)$ (why?)

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  - $\forall z(xz \in L \Leftrightarrow yz \in L)$. Then $x \sim_L y$ by definition of $\sim_L$
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  - That is, $x \equiv y \Rightarrow x \sim_L y$
- Since $\equiv$ is an *arbitrary* right congruence refining L, this holds for *every* such relation: each one refines $\sim_L$. Combined with $\sim_L$ itself being a right congruence refining L, $\sim_L$ is the coarsest such relation.

Myhill-Nerode Theorem

The following are equivalent :

1. L is regular
2. there exists a Myhill-Nerode relation $\sim$ for L
3. $\sim_L$, the relation defined (earlier) from L is of finite index (that is, induces finitely many equivalence classes).

Show $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$. each step reuses things we have shown already.

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- Then $\sim_A$ is a right congruence refining L with finitely many classes (shown earlier).
- Then $\sim_A$ is a Myhill-Nerode relation for L, witnessing (2).

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- Since $\sim$ has finite index, so does $\sim_L$.

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- So L is regular.

Table filling collapsed automaton and the Myhill-Nerode automaton

Table-filling and the Myhill-Nerode DFA

Assume $A = (Q, \Sigma, \delta, q_0, F)$ is a DFA for L which has been collapsed by the table-filling algorithm. Recall the equivalence computed by the table-filling algorithm

$$p \approx q \Leftrightarrow \forall x \in \Sigma^*(\hat{\delta}(p, x) \in F \Leftrightarrow \hat{\delta}(q, x) \in F)$$

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- Recall that we can define $\sim_A$ from A: $x \sim_A y \Leftrightarrow \hat{\delta}(q_0, x) = \hat{\delta}(q_0, y)$ as the Myhill-Nerode relation constructed from A.
- Recall that whenever L is regular, $\sim_L$ is the coarsest Myhill- Nerode relation we can define for L
- We show that $\sim_L$ and $\sim_A$ are the same

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- $\Leftrightarrow x \sim_A y$

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  Show that $\sim = \sim_{A_\sim}$.
- DFA B for language L to Myhill-Nerode relation $\sim_B$, back to DFA $A_{\sim_B}$.
  Show that B is isomorphic to $A_{\sim_B}$.

Putting it together

Two different routes to a minimal DFA for L:

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(B) Start from L. Define $\sim_L$ directly via residuals. Build $B_{\sim_L}$ whose states *are* the equivalence classes of $\sim_L$.

These are the same : once A has been collapsed by table-filling, $\sim_{A_{min}} = \sim_L$ exactly. Since A_{min}'s states are the classes of $\sim_{A_{min}}$, and $B_{\sim_L}$'s states are the classes of $\sim_L$, and these are the same classes, so A_{min} is isomorphic to $B_{\sim_L}$.